# Audible

Data Cleaning & Transformation Using Power Query

audible
an amazon company

# Audible – Company Overview

Audible is a leading digital audiobook and spoken-word entertainment platform owned by Amazon.

It provides audiobooks, podcasts, and exclusive audio content through a subscription-based model.

Users can purchase or stream content via the Audible app and listen anytime, anywhere.

The company generates revenue mainly through monthly memberships and audiobook sales.

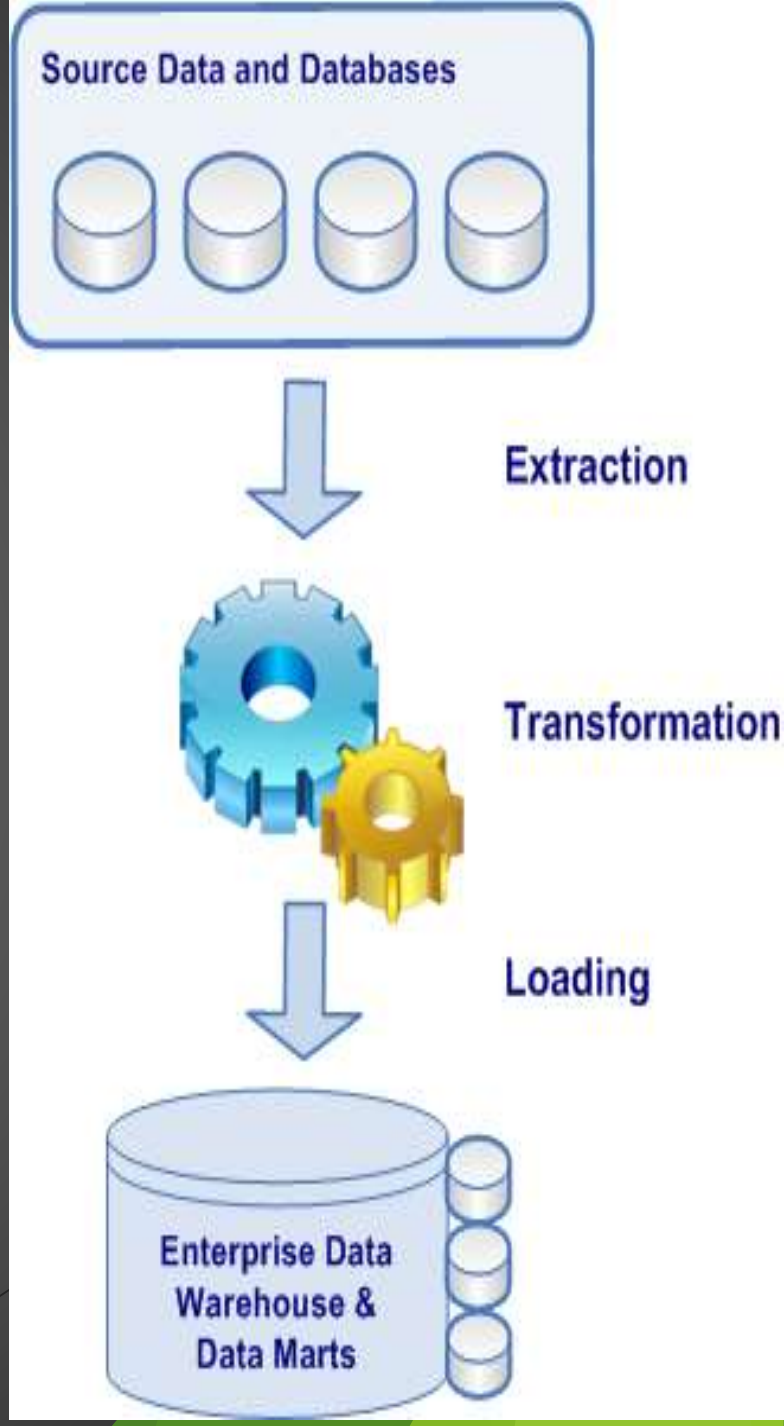
# INTRODUCTION

This project focuses on cleaning and transforming raw data using Microsoft Excel Power Query.

The dataset contained missing values, inconsistent formats, and incorrect data types that made it unsuitable for analysis.

Using ETL techniques, the data was cleaned, standardized, and structured into an analysis-ready format.

The goal was to improve data quality, accuracy, and reliability for better decision-making.



# Problem Statement

- **Inconsistent Data Formats**

Same data stored in different formats

Caused sorting and filtering errors

- **Non-Numeric Values in Numeric Columns**

Numbers stored as text or with symbols

Prevented accurate calculations

- **Missing / Null Values**

Blank entries in key columns

Affected data reliability

- **Duplicate Records**

Repeated audiobook entries

Led to incorrect totals and averages

- **Incorrect Date/Time Formats**

Dates stored as text

Duration not in proper time format

# Project Objective

- **Clean and Standardize Raw Data**

Removed errors, duplicates, and inconsistencies to ensure uniform formatting across the dataset.

- **Transform Data into Structured Format**

Converted unorganized raw data into a well-defined tabular structure suitable for analysis.

- **Ensure Correct Data Types**

Assigned appropriate data types (text, number, date, duration) to enable accurate calculations.

- **Improve Data Quality and Reliability**

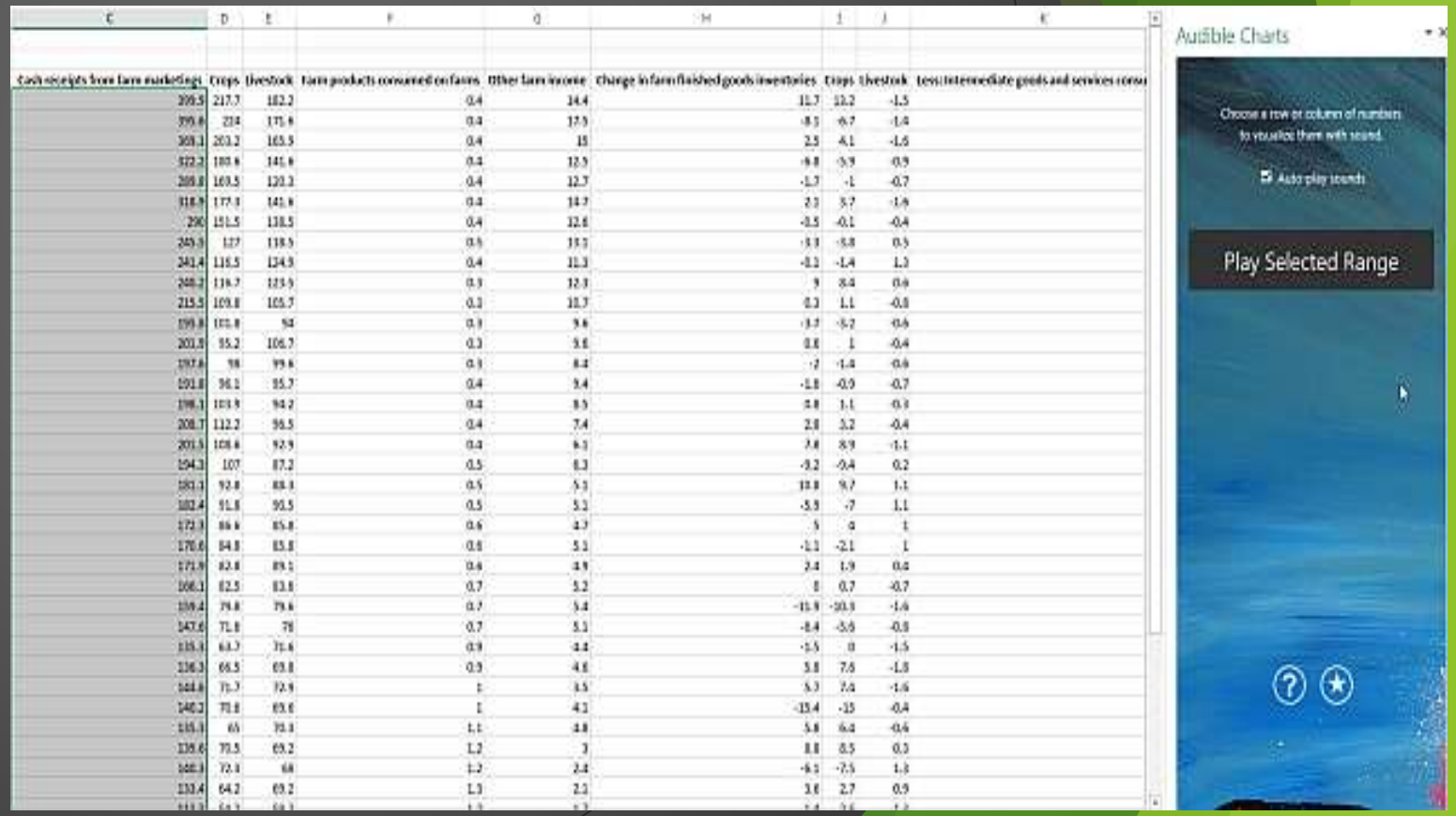
Enhanced accuracy and consistency to ensure trustworthy analytical results.

- **Prepare Dataset for Analysis & Reporting**

Created an analysis-ready dataset that can be directly used for dashboards and business reports.

# Data Overview

- **Source**: *Audible dataset (CSV file)*
- Total Rows & Columns (mention actual numbers from your dataset)
- **Key Columns**:
- *Book Title*
- *Author*
- *Rating*
- *Price*
- *Duration*
- *Release Date*



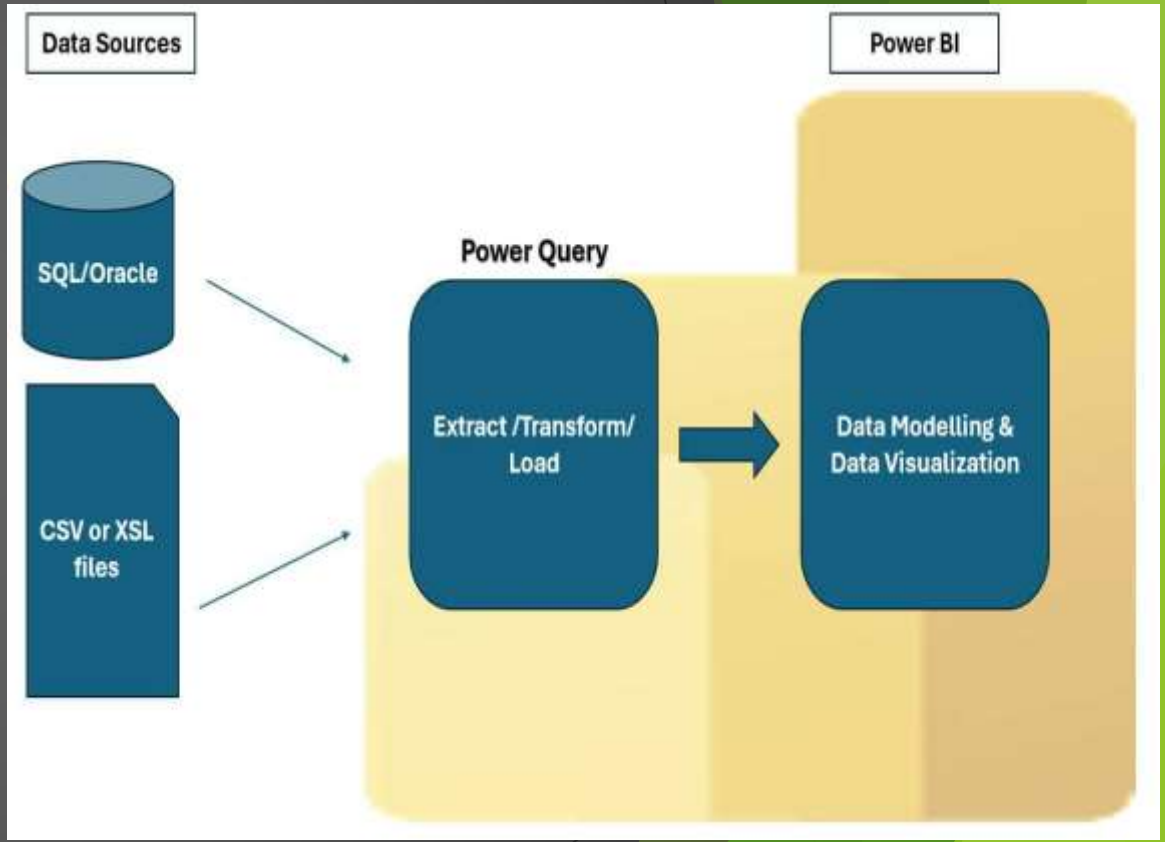
# Tool used-Power query

- **What is Power Query?**

1. A data transformation and ETL tool in Excel
2. Allows data cleaning without modifying source file
3. Uses M language in backend

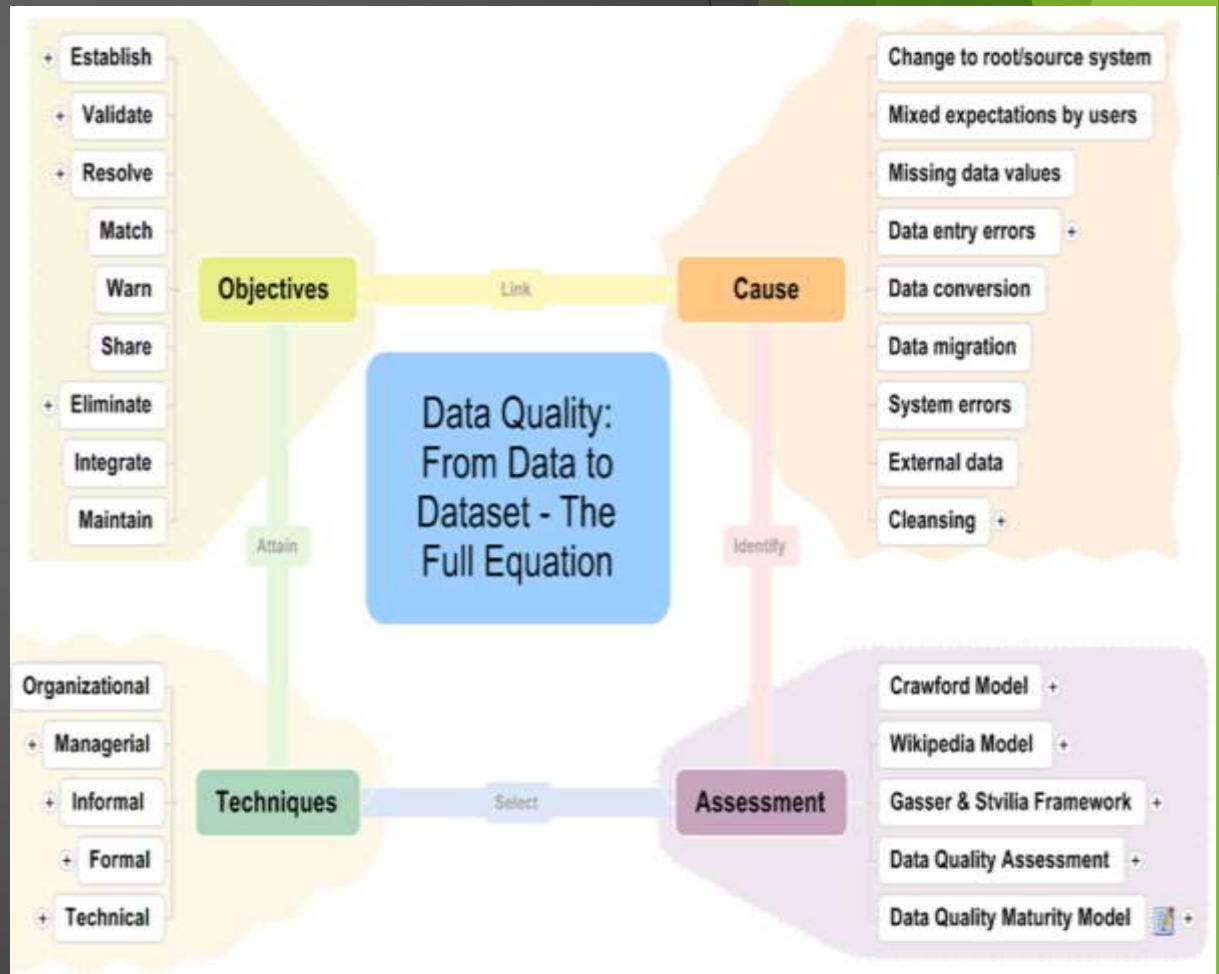
- **Why Power Query?**

1. Automated repeatable steps
2. Handles large datasets
3. Advanced transformation capabilities



# Data Issues Identified

- Price column contained text values
- Duration column in text format (e.g., "5 hrs 30 mins")
- Ratings stored as text
- Missing values in some columns
- Inconsistent date formats



# Data Cleaning Steps Performed

- **Remove Duplicates**

Use Remove Rows → Remove Duplicates on identifying key columns (Title + Author + Release Date).

- **Handle Missing Values**

Replace nulls with domain-appropriate defaults, or filter rows for manual review.

- **Change Data Types**

Price → Decimal, Ratings → Decimal, Duration → Duration, Date → Date. Validate with errors view.

- **Text Cleaning**

Trim, Clean, and Proper Case for titles/authors to ensure consistent joins and grouping

# Data Transformation Steps

- **Split & Extract**

Split Duration into hours/minutes; extract numeric portion from price and rating text.

- **Calculated Columns**

Create Total Minutes, Price Numeric, and Age From Release for easier analysis.

- **Merge & Rename**

Merge columns when needed and rename fields for reporting clarity

# Data Quality Improvements & Next Steps

**Numeric Consistency**
100% numeric values in price and ratings- enables accurate calculations

**Standardised Date**
Uniform date format supports trend analysis and time-based filtering.

**Reduced Manual Effort**
Automated Power Query steps reduce repetitive cleaning and human error.

**Suggested Next Steps**
Document queries, parameterize where required (currency, culture), and schedule refresh for updated CSVs.

# Business Impact

- **faster Reporting**

Automated Power Query steps reduce data preparation time, enabling quicker report generation.

- **Better Decision-Making**

Accurate and standardized data ensures reliable insights for informed business decisions.

- **Reduced Human Error**

Automation minimizes manual intervention, decreasing the chances of calculation and formatting mistakes.

- **Reusable Transformation Steps**

Power Query saves all cleaning steps, allowing the same process to be applied to new data easily.

# What I Learned

- Importance of clean data
- Power of ETL process
- Practical use of Power Query transformations
- Automation advantage over manual Excel cleaning

# Conclusion

Raw datasets often contain errors, inconsistencies, and missing values that must be cleaned before analysis.

Well-structured and validated data ensures reliable analysis and better decision-making

Power Query automates data cleaning and transformation processes efficiently with repeatable steps